

Programme	B. Sc. Computer Science				
Course Title	Computer Networks				
Type of Course	Major				
Semester	IV				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Knowledge in Computer Organization and Architecture. 2. Knowledge in Operating System.				
Course Summary	This course covers the concepts of data communication and computer networks. It comprises of the study of the standard models for the layered protocol architecture to communicate between autonomous computers in a network and also the main features and issues of communication protocols for different layers. Topics covered comprise of introduction to OSI and TCP/IP models also.				

Sl. NO:	Course Outcome	Cognitive level *	Knowledge category #	Evaluation Tools used
CO1	To understand the fundamentals of computer networks including concepts like data communication ,network topologies and the reference models	U	C	Instructor-Create Exams or Quiz
CO2	Proficiency in Transmission Media and Multiplexing Techniques:	A	P	Discussions and Quizzes
CO3	To familiarise with the common networking protocols and standards	U	F	Instructor created exams or Home assignments
CO4	Describe ,analyse and compare different data link, network and transport layer protocols	A, E	P	Discussions, Quizzes
CO5	Design/implement data link and network layer protocols in simulated networking environment	Ap	P	Viva Voce Observation of practical skills

CO6	To understand the need of various Application layer protocols	U	M	Instructor Created -Exams, Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Module	Unit	Content	Hrs	Marks
I	Introduction to Computer networks and Network models		12	17
	1	<i>Types of computer networks, Internet, Intranet, Network topologies, Network classifications.</i>	2	
	2	<i>Network Architecture Models: Layered architecture approach, OSI Reference Model, TCP/IP</i>	2	
	3	<i>Physical Layer: Analog signal, digital signal, Analog to Digital, Digital to Analog, maximum data rate of a channel transmission</i>	4	
	4	<i>Transmission media (guided - unguided transmission media)</i>	2	
	5	<i>multiplexing (frequency division multiplexing, time division multiplexing, wavelength division multiplexing)</i>	2	
II	Data Link Layer		11	18
	6	<i>Data link layer services, error-detection Types of errors, Single bit error and Burst error, Vertical redundancy check (VRC), longitudinal redundancy Check (LRC), Cyclic Redundancy Check (CRC), Check sum Error correction - Single bit error correction, Hamming code</i>	2	
	7	<i>Error correction techniques, error recovery protocols (stop and wait, go back n, selective repeat),</i>	3	
	8	<i>multiple access protocols, (TDMA/FDP, CDMA/FDD/CSMA/CD, CSMA/CA),</i>	2	
	9	<i>Datalink and MAC addressing, Ethernet, Polling</i>	1	
	10	<i>IEEE Standards- Wireless LANS, Ethernet, Bluetooth</i>	3	
III	Network layer		11	18
	11	<i>Networking and Internetworking devices - Repeaters, Bridges, Routers, Gateways, Firewall</i>	2	
	12	<i>Logical addressing - IPv4 & IPv6 addresses, Network Address Translation (NAT), Internet protocols, internetworking, Datagram,</i>	2	

	13	<i>Transition from IPv4 to IPv6</i>	1	
	14	<i>Address Mapping-Error reporting and multicasting - Delivery,</i>	2	
	15	<i>Forwarding and Routing algorithms, Distance Vector Routing,</i>	2	
	16	<i>Link State Routing. Dijkstra</i>	2	
IV	Transport Layer and Application layer		11	17
	17	<i>Transport layer, Process-to-process Delivery: UDP, TCP</i>	2	
	18	<i>Congestion control and Quality of Service,</i>	2	
	19	<i>Domain Name Systems-Remote Login, Email</i>	2	
	20	<i>FTP, WWW, HTTP</i>	2	
	21	<i>Introductory concepts on Network management & Mail transfer: SNMP,</i>	2	
	22	<i>SMTP</i>	1	
V	Hands-on Computer Networks: Practical Applications,		30	
	1	Lab 1: identifying Networking Hardware components(Jacks, Cables, Tools) Lab 2 IP address - configuring. Lab3. crimping Lab 4: Configuring network host - setting hostname - assigning IP address Lab 5: configuring the Network Interface card – Lab 6: Setup a Wired LAN with more than two systems Lab 7:Setup a Wireless LAN with more than two systems Lab 8: Setting up Internet services File Transfer Protocol(FTP), Lab 9: Simple Mail Transfer Protocol(SMTP) and Post Office Protocol(POP) Lab 10: Setting up Intranet Services - Network File System(NFS),	20	
	2	Case study	3	

	3	Capstone (/Course) Project: Build a practical application using Wired Network	7	
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References:

1. Behrouz A Forozan, *Introduction to Data Communications & Networking*, TMH
2. Andrew S. Tanenbaum, *Computer Networks*, PHI
3. William Stallings, *Data and Computer Communications*, VIIth Edition, Pearson Education